JAWAHARLAL NEHRU NEW COLLEGE OF ENGINEERING

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ROS 2 URDF OF 4-DOF INDUSTRIAL ARM

Project Report

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ROS Version:	ROS 2 Humble
Operating System:	Ubuntu 22.04.5 LTS
Package:	arm_4dof_description

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Abstract

This project presents the design and implementation of a 4-DOF industrial robotic arm using ROS 2 and URDF.

The robotic arm consists of a base link and four arm links connected through four revolute joints. Each joint is defined with its parent link, child link, axis, and motion limits. Visual and collision geometries are included for the robot.

The URDF model was validated using the `check_urdf` tool. The robot was visualized in RViz2 using `robot_state_publisher` and `joint_state_publisher_gui`. The TF tree was also generated and verified to confirm the connectivity of all robot links.

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Chapter 1

Introduction

URDF, or Unified Robot Description Format, is an XML-based format used in ROS to describe the structure and kinematic relationships of a robot.

In this project, a 4-DOF industrial robotic arm was designed and implemented using ROS 2 Humble.

The basic robot contains:

- One base link
- Four arm links
- Four revolute joints
- Visual geometry
- Collision geometry
- Joint limits

The completed robot was tested in RViz2 and its TF structure was verified.

Chapter 2

Objectives

The objectives of this project are:

1. Design a 4-DOF industrial robotic arm using URDF.
2. Define all robot links correctly.
3. Define the four revolute joints.
4. Specify parent and child links.
5. Specify appropriate joint axes.
6. Define joint limits.
7. Add visual and collision geometry.
8. Validate the URDF model.
9. Visualize the robot in RViz2.
10. Verify the TF tree.

Chapter 3

Software and Tools

Table 3.1: Software and Tools Used

Tool	**Version / Purpose**
Ubuntu	22.04.5 LTS
ROS 2	Humble
URDF	Robot description
RViz2	Robot visualization
robot_state_publisher	TF publishing
joint_state_publisher_gui	Joint state GUI
tf2_tools	TF tree generation
check_urdf	URDF validation

Chapter 4

Robot Structure

The robot consists of five links:

```
base_link
link1
link2
link3
link4
```

The robot contains four revolute joints:

```
joint1
joint2
joint3
joint4
```

The complete basic structure is:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
```

Chapter 5

URDF Links

5.1 Base Link

The base of the robotic arm is represented by `base_link`.

The base uses a cylindrical geometry.

- Radius: 0.30 m
- Height: 0.10 m

5.2 Link 1

The first arm link is represented by `link1`.

- Radius: 0.12 m
- Length: 0.60 m

5.3 Link 2

The second arm link is represented by `link2`.

- Radius: 0.10 m
- Length: 0.50 m

5.4 Link 3

The third arm link is represented by `link3`.

- Radius: 0.08 m
- Length: 0.40 m

5.5 Link 4

The fourth arm link is represented by `link4`.

- Radius: 0.06 m
- Length: 0.30 m

Chapter 6

Revolute Joints

Four revolute joints are used in the robotic arm.

Table 6.1: Joint Configuration

Joint	**Parent**	**Child**	**Axis**
joint1	base_link	link1	Z
joint2	link1	link2	Y
joint3	link2	link3	Y
joint4	link3	link4	Y

6.1 Joint Limits

Table 6.2: Joint Limits

Joint	**Lower**	**Upper**	**Effort**	**Velocity**
joint1	-3.14	3.14	10	1
joint2	-1.57	1.57	10	1
joint3	-1.57	1.57	10	1
joint4	-3.14	3.14	10	1

Chapter 7

URDF Validation

The URDF was validated using:

```
check_urdf urdf/arm_4dof.urdf
```

The result was:

```
robot name is: arm_4dof
---------- Successfully Parsed XML ---------------
root Link: base_link has 1 child(ren)
    child(1): link1
        child(1): link2
            child(1): link3
                child(1): link4
```

This confirms that the URDF was successfully parsed and that the links form a valid connected chain.

Chapter 8

RViz2 Visualization

The robot was loaded into RViz2 using `robot_state_publisher`.

The fixed frame was configured as:

```
base_link
```

The RobotModel display was enabled using the `robot_description` topic.

The four joint sliders were used to verify the movement of the four revolute joints.

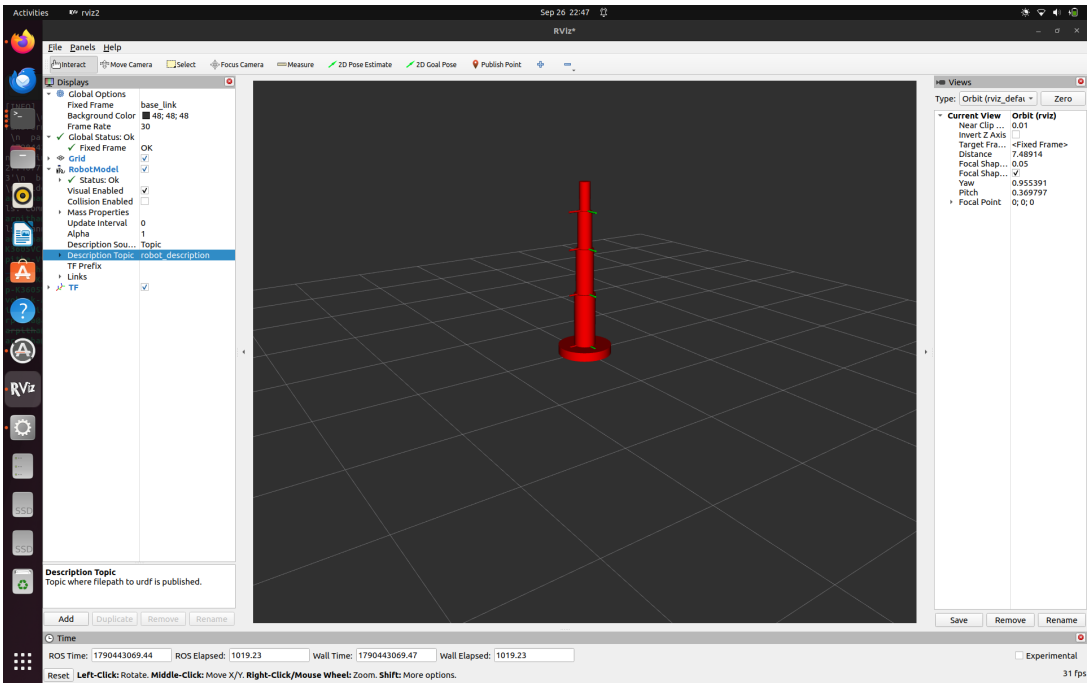


Figure 8.1: Basic 4-DOF industrial robotic arm in RViz2

Figure 8.1 shows the basic 4-DOF industrial robotic arm visualized in RViz2.

Chapter 9

TF Tree Verification

The TF tree was generated using:

```
ros2 run tf2_tools view_frames
```

The expected TF structure is:

```
base_link
    |
   link1
    |
   link2
    |
   link3
    |
   link4
```

The generated TF tree confirms that the four arm links are connected to the base.

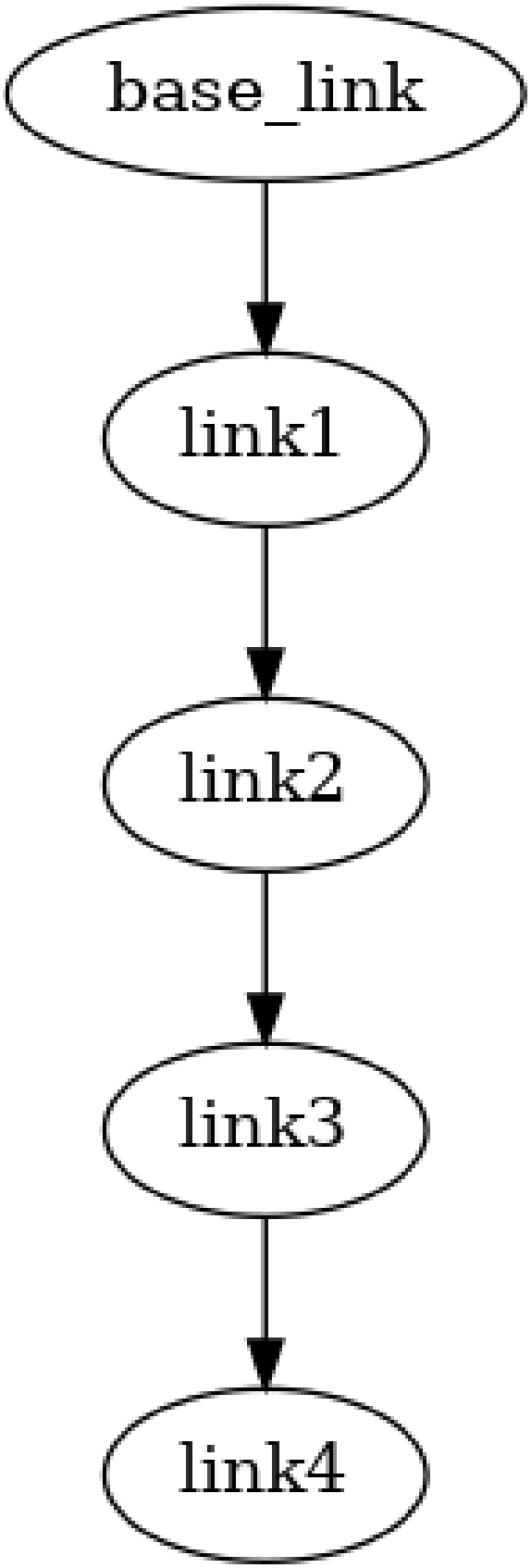


Figure 9.1: TF tree of the basic 4-DOF industrial robotic arm

Chapter 10

Conclusion

A 4-DOF industrial robotic arm was successfully designed using ROS 2 and URDF.

The robot contains four revolute joints with defined axes, limits, parent links, and child links.

Visual and collision geometries were included. The URDF was successfully validated using `check_urdf`.

The robot was visualized in RViz2 and the four joints were tested using the joint state GUI.

The TF tree was generated and verified, confirming the connectivity of the complete basic robot structure.